

The Exact Finite- n σ -MGF for S_n

Proof, orbit interpretation, and matrix verification

Computational proof companion

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Abstract

For the permutation representation $V = \mathbb{C}^n$, this note proves the exact cycle-type formula for $\mathcal{M}_{S_n}$, identifies its coefficients with vector partitions, and explains the stable range. The accompanying marimo notebook constructs all permutation matrices and compares a full Reynolds-projector rank with both formulas. All representative tests passed.

1 Molien coefficient as an invariant dimension

For $\tau = (\tau_1, \dots, \tau_s)$ let $W_\tau = \bigotimes_b \text{Sym}^{\tau_b}(\mathbb{C}^n)$. Since

$$\det(I - tg)^{-1} = \sum_{q \geq 0} \chi_{\text{Sym}^q V}(g) t^q,$$

the coefficient of m_τ in the generalized Molien series is

$$[m_\tau] \mathcal{M}_{S_n} = \frac{1}{n!} \sum_{g \in S_n} \chi_{W_\tau}(g) = \text{tr} \left(\frac{1}{n!} \sum_{g \in S_n} \rho_{W_\tau}(g) \right) = \dim W_\tau^{S_n}.$$

The middle operator is the Reynolds projector.

2 Exact cycle-type formula

Write a cycle type as $\lambda = (1^{m_1} 2^{m_2} \dots) \vdash n$ and set

$$z_\lambda = \prod_{d \geq 1} d^{m_d} m_d!.$$

A d -cycle has all d th roots of unity as eigenvalues, so $\det(I - tP_d) = 1 - t^d$. A permutation of type λ therefore contributes

$$\prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}.$$

The conjugacy class has size $n!/z_\lambda$, proving

$$\boxed{\mathcal{M}_{S_n} = \sum_{\lambda \vdash n} \frac{1}{z_\lambda} \prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}}.$$

This statement is exact for every n and in every degree; it is not a stable approximation.

3 Exact coefficient formula

A monomial-basis vector of W_τ is encoded by a labeling

$$\ell : \{1, \dots, n\} \longrightarrow \mathbb{Z}_{\geq 0}^s, \quad \sum_{j=1}^n \ell(j) = \tau$$

coordinatewise. Two labelings lie in the same S_n -orbit exactly when the multiplicities $a_{\mathbf{v}} = \#\{j : \ell(j) = \mathbf{v}\}$ agree. Since the dimension of the invariant space of a permutation module equals its number of orbits,

$$[m_\tau] \mathcal{M}_{S_n} = \# \left\{ (a_{\mathbf{v}})_{\mathbf{v} \geq 0} : a_{\mathbf{v}} \geq 0, \quad \sum_{\mathbf{v}} a_{\mathbf{v}} = n, \quad \sum_{\mathbf{v}} a_{\mathbf{v}} \mathbf{v} = \tau \right\}.$$

Equivalently, this is the number of vector partitions of τ into at most n nonzero parts; zero parts fill the remaining slots.

Marking the number of labels with u gives

$$\sum_{n \geq 0} u^n \mathcal{M}_{S_n} = \prod_{\mathbf{v} \in \mathbb{Z}_{\geq 0}^{(N)}} (1 - ut^{\mathbf{v}})^{-1}.$$

The exponential formula for permutation cycles gives the equivalent form

$$\sum_{n \geq 0} u^n \mathcal{M}_{S_n} = \exp \left(\sum_{d \geq 1} \frac{u^d}{d} \prod_i \frac{1}{1 - t_i^d} \right).$$

4 Stable range

Separate the zero label. Every nonzero vector has total degree at least one, so a vector partition of τ uses at most $|\tau|$ nonzero parts. If $n \geq |\tau|$, zeros can always be added and the coefficient stabilizes:

$$[m_\tau] \mathcal{M}_{S_n} = [m_\tau] \prod_{\mathbf{v} \neq 0} (1 - t^{\mathbf{v}})^{-1}.$$

When $n < |\tau|$, the exact finite- n formula simply removes decompositions requiring too many nonzero parts.

5 Verification method and evidence

For each $g \in S_n$, the notebook creates its permutation matrix, compresses $g^{\otimes q}$ to an orthonormal occupancy basis of $\text{Sym}^q(V)$, tensors the selected blocks, and averages the resulting matrices. The projector rank is compared independently with the symmetric-power character average, cycle-type coefficient extraction, and a dynamic-programming vector-partition count.

Case	$\dim W_\tau$	rank	character	cycle	orbit
$S_2, \tau = (3)$	4	2	2	2	2
$S_3, \tau = (2, 1)$	18	4	4	4	4
$S_4, \tau = (2, 2)$	100	9	9	9	9
$S_5, \tau = (2, 1)$	75	4	4	4	4

The largest projector here is 100×100 and averages 24 explicitly constructed matrices. Idempotence residuals range from 2.2×10^{-16} to 1.1×10^{-15} . These cases cover degree above the stable threshold, multiblock tensor products, and two different values of n for the same τ .

Conclusion. The cycle-index and orbit formulas are two descriptions of the same invariant dimension, and the explicit matrix calculations agree with both in every tested case.